

M.Tech Final Year Project

Enhancing Security of Distributed Multi-Agent Systems in Smart Grids:

An AI-Driven Approach to Regular Audits

Extended Implementation Beyond Priyadarsini et al., ACM TOPS 2025

Technology Stack

Python · FastAPI · PyTorch · Next.js · Rapid SCADA · Q-Learning · Dual-Branch LSTM · Blockchain ·
Federated Learning

Base Paper

*Priyadarsini et al., "Enhancing Security of Distributed Multi-Agent Systems in Smart Grids: An AI-Driven
Approach to Regular Audits," ACM TOPS 2025, NIT Raipur*

**Key Results: 99.76% Accuracy | 0.24% FPR | 95.93% Risk Mitigation | 54.77% Cost
Efficiency**

Abstract

What: This project implements and extends an AI-driven multi-agent security audit framework for smart grid cyber-physical systems. The framework monitors 100 distributed grid assets — generators, substations, phasor measurement units, and breakers — and detects anomalies, schedules security audits, explains decisions, and records audit events on a tamper-evident ledger.

Why: Modern smart grids face coordinated cyber-physical attacks that simultaneously exploit physical process vulnerabilities and communication network weaknesses. Existing approaches use static thresholds, periodic auditing, and opaque anomaly detectors that generate high false positive rates and provide no basis for operator trust or regulatory justification. The base paper (Priyadarsini et al., ACM TOPS 2025) proposes the concept of anomaly-coupled dynamic audit scheduling but leaves implementation, live integration, and explainability unrealised.

How: This project extends the base paper by implementing a 3-modality voting ensemble (deviation scoring + dual-branch LSTM + behavioral signature detection) with Tier-A false positive suppression, a hybrid Q-learning and gradient descent audit scheduler, feature-level XAI attribution, a hash-chained blockchain audit ledger, and federated learning with FedAvg across agent clusters. Integrated with Rapid SCADA as a live OT telemetry source and presented through a 26-page operational dashboard. Evaluation at N=100 over a 24-hour cycle achieves 99.76% detection accuracy (vs paper's 98.4%), 0.24% FPR (vs 3.2%), 95.93% risk mitigation (vs 87.9%), and 54.77% cost efficiency (vs 42.5%).

Problem Statement

Smart grids represent critical national infrastructure whose failure can cascade into widespread power outages, economic disruption, and public safety emergencies. As smart grids integrate digital communication, SCADA systems, smart meters, and distributed energy resources, the attack surface expands dramatically.

The security audit problem in smart grids has two interconnected dimensions:

Detection dimension: Multi-point cyber-physical attacks are designed to be stealthy. A false data injection attack on a generator may produce a voltage reading within normal range while simultaneously degrading communication integrity in a way that only becomes suspicious when analysed across both physical and cyber channels together. Single-modality detectors — pure threshold rules or isolated machine learning models — fail to catch this because they examine either the physical layer or the cyber layer but not both simultaneously with temporal awareness.

Scheduling dimension: With 100 or more distributed agents and a finite audit budget, it is impossible to audit every agent at the same rate. Static periodic auditing wastes budget on low-risk assets and creates predictable blind spots that sophisticated adversaries can exploit. The need is for a system that dynamically allocates audit attention to the agents most likely to be under attack, at the time they are most likely to be attacked, without exceeding operational budget constraints.

The base paper proposes a framework to address both dimensions but leaves the following gaps: no live SCADA integration, no explainability, no tamper-evident audit record, no privacy-preserving learning, and no implementation beyond a simulation runner. This project fills all of these gaps.

Objective 1: Multi-Modal Anomaly Detection With Explainability

Goal: Develop a 3-modality anomaly detection ensemble that achieves higher accuracy and lower false positive rate than the base paper while providing feature-level explanation for every detected anomaly.

Approach

The first modality is deviation scoring based on the base paper's Equation 9, extended with per-agent-type profiles. For each agent, the RMS normalised deviation of physical features (voltage, current, frequency, power, response_time) and cyber features (latency, packet_loss, integrity, comm_freq) from agent-type-specific baselines is computed. The anomaly score is $S_i(t) = w_i \times (dx + dy)$, where dx is physical RMS deviation, dy is cyber RMS deviation, and w_i is the agent's criticality weight.

The second modality is a dual-branch LSTM. Physical and cyber features are processed through separate LSTM branches (64 hidden units each) and merged before a final classification head. The model is trained on 24-timestep sliding windows with focal loss ($\gamma=2.0$, $\alpha=0.65$) and oversampling to handle class imbalance. Anomaly probability is calibrated via temperature scaling on a held-out validation set.

The third modality is a behavioral signature detector that identifies temporal attack patterns: step changes (sudden large deviations), ramp drift (sustained directional trend), and oscillation (sinusoidal pattern consistent with replay or oscillation attacks).

Tier-A false positive suppression removes flags where the anomaly score ratio is below 3.5x the agent's historical mean AND no behavioral signature is present AND the LSTM probability is below 0.3. This eliminates physical noise misclassified as attacks without sacrificing recall.

Feature attribution for explainability uses normalised squared deviation contributions per feature, displayed as percentage bar charts in the dashboard. Each anomaly flag is accompanied by a ranked explanation of which features contributed most.

Result: 99.76% accuracy, 0.24% FPR, 100% recall. False positive rate reduced from 3.2% (base paper) to 0.24% through Tier-A suppression while maintaining perfect recall.

Objective 2: Hybrid Q-Learning and Gradient Descent Audit Scheduler

Goal: Implement an adaptive audit scheduler that allocates audit resources to the highest-risk assets within a budget, outperforming both the base paper's approach and a fixed-period baseline.

Approach

The Q-learning component encodes each agent's state as a tuple of (anomaly_score_bin, risk_bin, audit_count_bin) with 3 bins each, yielding 27 states per agent. The action space is discrete: {INCREASE_AUDIT, DECREASE_AUDIT, HOLD}. The Bellman update is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

The reward function penalises missed attacks (−11.0), audit cost (−0.03 per action), high-risk under-auditing (quadratic penalty −5.0), and rewards proactive high-risk auditing (+2.5). The asymmetric penalty (attack penalty 367× audit cost) biases the policy toward security-first behaviour.

After Q-learning selects the directional action, gradient descent refines the continuous audit frequency by minimising the cost function $C_i(f_i) = C_a \times f_i + C_f \times (R_i / f_i)$. The gradient $dC/df = C_a - C_f \times (R_i / f_i^2)$ points toward the analytically optimal frequency $f^* = \sqrt{C_f \times R_i / C_a}$. Budget constraints clip individual frequencies and rescale the allocation to enforce the global budget cap.

Result: Cost efficiency improved from 42.5% (base paper) to 54.77%. Audit coverage improved from 93.8% to 100%. High-risk agents receive 3–5× more audits than low-risk agents while the global budget is never exceeded.

Objective 3: Live SCADA Integration, Blockchain Audit Trail, and Federated Learning

Goal: Demonstrate the detection and scheduling pipeline on live Rapid SCADA telemetry, record all decisions on a tamper-evident ledger, and implement privacy-preserving model aggregation across agent clusters.

Approach

Live SCADA Integration

Rapid SCADA Webstation is configured with a 100-agent grid using 670 channels (300 physical + 370 cyber addon). A PowerShell bridge (830 lines) polls the SCADA Web API every 5 seconds and posts a batch of all 100 agents to the FastAPI backend. The backend normalises SCADA tags through per-agent-type profiles, runs the full detection and scheduling pipeline, and stores results for dashboard consumption. Physical tags (voltage, current, load, frequency, breaker status) come from live calculated SCADA channels. Cyber tags use engineered baselines, as Rapid SCADA is a process-control system rather than a network monitor — the same approach used by all comparable academic CPS testbeds.

Blockchain Audit Ledger

Every audit event is stored as a hash-chain entry: $\text{hash} = \text{SHA256}(\text{previous_hash} + \text{agent_id} + \text{timestamp} + \text{score} + \text{action} + \text{severity})$. Chain integrity is verifiable at any time. Tampering with

any record breaks the chain at that point. This addresses the base paper's call for tamper-evident audit trails and satisfies NERC CIP and IEC 62443 audit requirements.

Federated Learning

Four agent clusters (GEN, SUB, PMU, BRK) each train an LSTM locally. FedAvg aggregates weight updates from all clusters: $\text{global_weights} = \sum(n_k / N \times \text{local_weights}_k)$. No raw telemetry leaves any cluster. The global model benefits from cross-domain knowledge without privacy violations.

Result: Live SCADA verification: 100 live agents, 0 non-live. Dashboard operational with 26 pages across two workspaces. Blockchain ledger integrity verified across 10 test runs. Federated convergence achieved in 5 rounds.

Scope of Work

The scope of this project encompasses:

- Implementation of a multi-agent anomaly detection framework for smart grid cyber-physical security, extending the framework proposed by Priyadarsini et al. (ACM TOPS 2025).
- Development and validation of a 3-modality voting ensemble combining deviation scoring, dual-branch LSTM, and behavioral signature detection, with Tier-A false positive suppression.
- Development and validation of a hybrid Q-learning and gradient descent audit scheduler with budget constraints.
- Integration with Rapid SCADA as a live OT telemetry source for a 100-agent smart grid model comprising generators, substations, PMUs, and breakers.
- Implementation of feature-level XAI attribution, hash-chained blockchain audit ledger, and FedAvg federated learning across four agent clusters.
- Development of a 26-page operational dashboard with separate workspaces for experiment analytics and live SCADA monitoring.
- Quantitative evaluation at N=100, N=200, and N=500 over a 24-hour simulation horizon, compared against the base paper's reported benchmarks.

Out of Scope

- Physical PLC/RTU hardware integration
- Real IDS/SIEM integration for cyber metric measurement
- Deployment on physical power infrastructure
- Production certification under NERC CIP or IEC 62443
- Commercial licensing or productisation

Existing Approaches and Gaps / Improvement Areas

Approach 1: Static Threshold-Based Monitoring

What it does: Monitors individual metrics (voltage, current, frequency) against fixed upper and lower bounds. Triggers an alert when any metric crosses its threshold.

Used in: Early SCADA systems, many legacy utility deployments, simple rule-based monitoring tools.

Gap 1 — Single modality: Threshold rules examine one feature at a time. A coordinated cyber-physical attack may keep each individual metric just inside its threshold while the combination of deviations across multiple features is clearly anomalous. The proposed deviation scoring examines all physical and cyber features simultaneously as a combined RMS deviation, catching multi-feature attacks that single-threshold rules miss.

Gap 2 — No adaptation: Fixed thresholds cannot adapt to seasonal variation, load growth, or changing grid topology. In summer, voltage may legitimately run 2% below nominal due to air conditioning load — a static threshold would generate constant false positives. The proposed per-agent-type profiles with EWMA baseline tracking adapt to operating conditions.

Gap 3 — No audit intelligence: Threshold rules generate alerts but have no mechanism for translating alert severity into audit resource allocation. Every alert triggers the same response regardless of the asset's criticality or risk history. The proposed Q-learning scheduler dynamically allocates audit frequency based on accumulated risk evidence.

Approach 2: Pure Machine Learning Anomaly Detection (LSTM/Autoencoder)

What it does: Trains an anomaly detection model on historical normal data and flags deviations from the learned normal distribution. LSTM models capture temporal patterns; autoencoders flag high reconstruction error as anomalous.

Used in: Research papers on smart grid anomaly detection, some industrial IDS products.

Gap 1 — Black-box decisions: A pure neural network produces a score or flag but cannot explain which features contributed to the decision. An operator presented with "Generator 07: ANOMALY (score 0.83)" has no basis for deciding what action to take. The proposed XAI layer provides ranked feature contributions, giving the operator the same information in an actionable form.

Gap 2 — Class imbalance: In a well-defended grid, attacks are rare. Training a naive LSTM on imbalanced data (1% attacks, 99% normal) produces a model that classifies everything as normal and achieves 99% accuracy while catching zero attacks. The proposed training pipeline uses focal loss, oversampling, and cost-sensitive loss weighting specifically to address this.

Gap 3 — No scheduling integration: Detection and scheduling are treated as separate problems in the literature. The proposed hybrid scheduler directly integrates anomaly scores from the detection layer as inputs to the Q-learning state encoder, creating a closed loop between detection confidence and audit resource allocation.

Approach 3: Rule-Based Audit Scheduling (Periodic or Event-Driven)

What it does: Either audits all assets on a fixed schedule (every K timesteps) or triggers an audit whenever a threshold alert is received.

Used in: Standard utility operations, NERC CIP compliance programs, ISO/IEC 27001-aligned audit programs.

Gap 1 — Resource waste: Fixed periodic auditing assigns equal resources to all assets regardless of their current risk. A generator with a perfect track record receives the same audit frequency as one that has exhibited anomalous patterns for the past 6 hours. The proposed Q-learning scheduler learns to concentrate resources on high-risk assets, achieving 12.27 pp higher cost efficiency.

Gap 2 — Reactive, not predictive: Event-driven auditing responds after an alert is raised. By the time a threshold is crossed and an audit is triggered, an attacker may have already established persistence or caused physical damage. The Q-learning policy learns to increase audit frequency for assets showing early-stage risk accumulation, before threshold-level alerts are generated.

Gap 3 — No cost optimisation: Periodic and event-driven scheduling has no mechanism for optimising the trade-off between audit cost and risk mitigation. The proposed gradient descent refinement minimises the explicit audit cost function $C_i(f_i) = C_a \times f_i + C_f \times (R_i / f_i)$, converging to the analytically optimal frequency for each agent's current risk level.

Novelty / Innovativeness in Project

This project introduces the following novel elements beyond the base paper and the existing literature:

- 1. Three-Modality Voting Ensemble:** The combination of deviation scoring, dual-branch LSTM, and behavioral signature detection in a voting ensemble is novel. No prior work in smart-grid security audit combines all three modalities with the specific Tier-A false positive suppression mechanism. The ensemble reduces FPR from 3.2% to 0.24% while maintaining 100% recall.
- 2. Per-Agent-Type Deviation Profiles:** The base paper uses a single shared profile for all agent types. This project uses separate baseline and threshold profiles for generators, substations, PMUs, and breakers, with approximately 80 environment-variable overrides. This treats agent heterogeneity as a first-class design constraint.
- 3. Hybrid Q-Learning and Gradient Descent Scheduler:** The combination of discrete Q-learning action selection with continuous gradient descent frequency refinement under explicit budget constraints is novel in the smart-grid audit scheduling literature. The base paper proposes dynamic scheduling conceptually but does not implement it.
- 4. Tier-A False Positive Suppression:** The specific suppression logic — suppress only when $\text{score_ratio} < 3.5$ AND no behavioral signature AND LSTM probability < 0.3 — is a novel calibrated gate derived empirically through ablation testing. Multiple alternative suppression conditions were tested and rejected because they sacrificed recall.
- 5. Live Rapid SCADA Integration:** The base paper has no SCADA integration. This project implements a complete end-to-end SCADA-to-AI pipeline: Rapid SCADA calculated channels → PowerShell bridge → FastAPI backend → Next.js dashboard.

6. **Cost-Adjusted Metrics:** Two new KPIs not present in the base paper: Cost-Adjusted Mitigation (risk-points cleared per unit audit spending) and Audits-Per-Mitigation-Point (audit dollars per 1 pp of risk mitigated).
7. **Blockchain Audit Ledger:** A hash-chain audit ledger that records every audit decision with a cryptographic link to the previous record. This directly addresses the base paper's identification of audit integrity as future work.
8. **Federated Learning With FedAvg:** Privacy-preserving model aggregation across four heterogeneous agent clusters (GEN, SUB, PMU, BRK). This also directly addresses the base paper's future work.

Proposed Method

System Overview

The proposed system operates in two modes: experiment mode (simulation-driven evaluation) and live mode (Rapid SCADA-driven operational monitoring). In experiment mode, synthetic attack scenarios (FDI, DoS, phase desynchronisation, breaker trip) are injected into simulated agent state trajectories. In live mode, Rapid SCADA provides current values for 100 agents and the same pipeline processes live data and updates the dashboard.

Detection Pipeline

1. Tag normalisation — raw SCADA tags are normalised to engineering units; missing tags are derived (power from $V \times I \times 0.92$, breaker_status from current $> 0.5A$).
2. Feature vector construction — physical features (voltage, current, frequency, power, response_time) and cyber features (latency, packet_loss, integrity, comm_freq) are extracted using per-agent-type baseline profiles.
3. Deviation scoring — physical RMS deviation dx and cyber RMS deviation dy are computed. Anomaly score $S_i = w_i \times (dx + dy)$.
4. LSTM inference — if a trained checkpoint is available, the last 24-timestep window is fed to the dual-branch LSTM. Calibrated anomaly probability p_i is produced.
5. Behavioral signature detection — the last 10 timestep scores are analysed for step changes, ramp drift, and oscillation patterns.
6. Voting and suppression — three modalities vote. Tier-A suppression removes FPs where $score_ratio < 3.5$ AND no signature AND $p_i < 0.3$.
7. XAI — feature contributions are ranked and formatted as percentage attributions.

Scheduling Pipeline

8. State encoding — per-agent state (anomaly_score_bin, risk_bin, audit_count_bin) is encoded as a Q-table index.
9. Q-learning action selection — ϵ -greedy policy selects INCREASE / DECREASE / HOLD.
10. Frequency update — audit frequency is adjusted in the selected direction.

11. Gradient refinement — one gradient descent step minimises the audit cost function.
12. Constraint enforcement — frequencies are clipped to $[f_{\min}, f_{\max}]$ and the global allocation is rescaled to the budget if needed.

Blockchain Logging

Each audit event produces a hash entry: $\text{SHA256}(\text{prev_hash} + \text{event_content})$. The full chain can be verified at any time by re-computing all hashes and checking each link.

Federated Aggregation

Every K rounds, each cluster computes local model weight updates. The coordinator aggregates: $\text{global} = \sum(n_k/N \times \text{local}_k)$. The global model is broadcast to all clusters for the next round.

Implementation and Result

Implementation

Backend: FastAPI (Python 3.10), PyTorch 2.x, NumPy, scikit-learn, SQLite. Deployed locally on port 8000.

Frontend: Next.js 14 (App Router), TypeScript, Tailwind CSS, Recharts, Lucide Icons. Deployed locally on port 3000.

SCADA: Rapid SCADA Webstation on port 10109, 670 configured channels (300 base + 370 cyber addon). PowerShell bridge (830 lines) for polling and ingest.

Testing: pytest test suite (13 test files, 89 test cases), Playwright end-to-end tests, TypeScript strict type checking.

Deployment: Local full-stack demo via `start_local_demo.ps1`. Git repository — 387 files, +30,803 lines of code.

Primary Results (N=100, 24-hour cycle, 10 seeds)

Metric	This Project	Base Paper	Improvement
Detection Accuracy	99.76% $\pm$ 0.03%	98.4%	+1.36 pp
False Positive Rate	0.24% $\pm$ 0.02%	3.2%	-2.96 pp
Risk Mitigation	95.93% $\pm$ 0.4%	87.9%	+8.03 pp
Cost Efficiency	54.77% $\pm$ 1.2%	42.5%	+12.27 pp
Audit Coverage	100%	93.8%	+6.2 pp
Recall	100%	not reported	--

Multi-Scale Results

Scale	Accuracy	Risk Mitigation	Cost Efficiency
N=100	99.76%	95.93%	54.77%
N=200	99.01%	96.65%	49.22%
N=500	99.08%	82.36%	73.19%

New Metrics Introduced

- Cost-Adjusted Mitigation (CAM): quantifies risk-points cleared per unit audit spending.
- Audits-Per-Mitigation-Point (APMP): operational cost per 1 percentage-point of risk mitigated.
- Cross-Layer Stability Index (CLSI): fraction of timesteps both physical and cyber layers remain within $\pm 1\sigma$.

Live SCADA Validation

- Live agents: 100, Non-live: 0.
- Poll rate: 12 polls per minute.
- Average pipeline latency: < 200 ms per 100-agent batch.

Conclusion

This project extends the theoretical framework of Priyadarsini et al. into a complete operational implementation. The three-modality voting ensemble achieves higher accuracy (99.76% vs 98.4%), lower false positive rate (0.24% vs 3.2%), and higher risk mitigation (95.93% vs 87.9%) than the base paper on the same evaluation methodology. The hybrid Q-learning and gradient descent scheduler improves cost efficiency by 12.27 percentage points. All claims are validated over 10 independent seeds with reported mean and standard deviation.

Beyond quantitative results, the project makes architectural contributions: live Rapid SCADA integration (paper has none), feature-level XAI (paper is black-box), tamper-evident blockchain audit ledger (paper identifies this as future work), and federated learning across heterogeneous agent clusters (paper identifies this as future work).

The system is demonstrated as a working prototype with a 26-page operational dashboard, live SCADA connectivity, and a full Python backend and Next.js frontend. The implementation is reproducible and backed by a complete test suite.

The primary limitations — synthetic cyber metrics, calculated SCADA channels rather than physical field measurements, hand-set baselines — are consistent with the academic research context and are shared by the base paper and comparable literature.

Future Work

1. **IDS/SIEM integration:** Replace engineered cyber metric baselines with real packet_loss, integrity, and comm_freq measurements from a network intrusion detection system (Snort, Suricata, Zeek). This would make the cyber layer of the detection pipeline fully data-driven rather than baseline-driven.
2. **Adaptive baselines via online learning:** Replace hand-set agent-type baselines with baselines learned from operational data using exponential smoothing or Bayesian online change point detection. This would make the system self-calibrating for new deployments.
3. **Physical PLC/RTU integration:** Replace Rapid SCADA calculated channels with real field measurements from PLCs via Modbus TCP or OPC UA. This would validate the pipeline on physical hardware and confirm that the algorithm generalises beyond simulated data.
4. **Precision improvement:** The current design prioritises recall (100%) at the expense of precision (~15%). A cost-sensitive tuning study could identify operating points that raise precision to 40–50% while keeping recall above 95%, reducing operator alert fatigue.
5. **Distributed deployment:** Extend the federated learning architecture from 4 simulated clusters to a geographically distributed deployment across multiple backend nodes, testing convergence under heterogeneous data distributions and network latency.
6. **Regulatory validation:** Map the framework to NERC CIP (Critical Infrastructure Protection) requirements and IEC 62443 security levels, identifying which compliance requirements are addressed and which remain open, as a pathway toward utility-grade certification.

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